

Software Requirements Specification for IBM AI powered consultancy simulation (IBM Consultancy 101)

Version 1.0

**Prepared by 09-IBM-AI-Powered Consulting Simulation — Team
1**

RMIT University Capstone

10/08/2026

Contents

Contents	1
Revision History	2
1.Introduction	3
1.1Purpose.....	4
1.2Problem Statement	4
1.3Document Conventions	4
1.4Definitions and Acronyms	5
1.5Intended Audience and Reading Suggestions.....	5
1.6Product Scope	6
1.7References	6
2.Overall Description.....	6
2.1Product perspective	6
2.2Product Functions	6
2.3User Classes and Characteristics	6
2.4Operating Environment	7
2.5Design and Implementation	7
2.6Assumptions and Dependencies	7
2.6Edge Cases and Exceptional Scenarios	8
3.Functional Requirements.....	8
3.1User Access and Onboarding	9
3.2Consulting World and Navigation.....	9
3.3Lead Discovery.....	9
3.4Cold Outreach	10
3.5Meeting Preparation	10
3.6AI powered Client Interaction	10
3.7Client Relationship and Meeting Outcomes.....	11
3.8Proposal and Contract.....	11
3.9Portfolio and Progress Tracking	12
3.10Feedback and Learning.....	12
3.11Client Personas	12

3.12Simulation Progression	13
----------------------------------	----

Revision History

Name	Date	Reason for changes	Version

--	--	--	--

1. Introduction

1.1 Purpose

This project is an AI powered consultancy simulation web application that allows graduates and trainees at IBM's consultancy practice to learn consulting skills such as cold outreach, emailing, and meeting preparation in a low-stakes environment, rather than the current practice of learning on the job, in front of real clients during real high-stake projects.

This Software Requirements Specifications (SRS) covers the complete IBM Consultancy 101 system as planned for a development throughout the three-sprint timeline. This document addresses functional and nonfunctional requirements for the core features including user onboarding, 2D world navigation, lead discovery and evaluation, cold outreach, meeting preparation, AI powered client engagement, engagement outcomes, portfolio tracking, and learning feedback.

This document is intended for use by the development team (Team 9), project stakeholders, and product owners to ensure a shared understanding of the system requirements and project scope.

1.2 Problem Statement

New graduate consultants and trainees entering IBM's consulting practice may have limited practical experience with activities such as identifying leads, conducting cold outreach, preparing for client meetings, handling client objections, and progressing an engagement toward a contract. Traditional training approaches may not provide sufficient opportunities to practise these situations in a realistic, low risk environment.

The application aims to address this gap by providing an interactive simulation in which trainees and graduate consultants can practise the consulting lifecycle and experience realistic client interactions without the consequences of making mistakes with real clients in real, high-stakes projects.

1.3 Document Conventions

This SRS is based off the IEEE 830 standard. The following conventions are used throughout this document:

Text formatting:

- **Bold** text indicates important terms, system names, or section headings
- *Italic* text is used for emphasis or to highlight specific concepts and technical terms
- "Quoted text" indicates user interface labels, button names, or exact system messages

Requirement Identification:

- Functional requirements are numbered using the format FR-X.Y.Z (e.g., FR-1.1.1)
- Non-functional requirements use the format NFR-X.Y.Z (e.g., NFR-1.1.1)
- User stories are referenced as US-X (e.g., US-1)
- Business requirements are numbered using the format BR-X.Y.Z (e.g. BR-1.1.1)
- Assumptions are numbered AS-X (e.g. AS-1)

Priority Classification:

- High (H): Core functionality required for MVP
- Medium (M): Important features that enhance user experience
- Low (L): Extension features and enhancements

Technical Conventions:

- System components are referenced in UPPERCASE (e.g., DATABASE, API, UI)
- User roles are capitalised (e.g., User, Admin)
- Technology stack components are specified as:
 - o React
 - o Motion
 - o Phaser
 - o Next.js
 - o Firebase

1.4 Definitions and Acronyms

LLM : Large Language Model

NPC: Non-Player Character

AI: Artificial Intelligence

SRS: Software Requirements Specification

1.5 Intended Audience and Reading Suggestions

This document is intended for a wide range of readers, including developers, project managers, client, and testers for this project. This offers the developers the comprehensive specifications required for the design, implementation, and integration of the application. Additionally, project managers will be able to use this document to plan timelines as well as track progress throughout the project. The functional and nonfunctional requirements outlined will provide testers with the basis for creating efficient and high quality tests to ensure the program functions appropriately.

All readers should begin with the introduction and focus on Section Two of the document entailing the overall description of the application, including perspective and important functions to help grasp the scope of the project. However, developers, project managers, and

testers for this application should also ensure that Section Three and Four is thoughtfully considered as these requirements will be crucial to the overall understanding of this project.

1.6 Product Scope

This is a web-based platform created to aid graduates and trainees at IBM to learn consulting skills before dealing with real clients. Its primary *goal* is to give IBM graduates and trainees consulting skills and confidence in these skills to equip them to take on real clients and projects. The main *benefit* of this application is that it runs on a single platform for the user to develop their skills in a low stake environment.

This project will include a 2D website game covering the full consulting loop, an LLM client persona, game state tracking portfolio progress and consulting practices based on IBM's practice.

This project will not support multiplayer, a full production deployment for IBM, voice acted dialogue, or native mobile builds, as decided with the Product Owner.

1.7 References

1. IEEE SA. (1998). IEEE 830-1998. Retrieved from <https://standards.ieee.org/ieee/830/1222/>

2. Overall Description

2.1 Product perspective

Add diagram of application

2.2 Product Functions

Add functions

2.3 User Classes and Characteristics

This primary users of this application are new Graduates and Trainees entering IBM's consultancy practice. These users are expected to have limited to no experience in professional consulting experience and may therefore require some guidance when

performing common consulting activities. Their technical experience may vary, but users are not expected to have or need advanced technological knowledge to use this application.

The primary goal is for the user to develop practical consulting skills and confidence before taking on real clients and projects. In this application, they are expected to explore elads, communicate with clients, prepare for meetings, appropriately respond to client interactions and progress engagements. They are to access this primarily via a desktop web browser during their individual training or learning sessions.

The application is designed primarily for users who **don't already possess extensive consulting experience**. Therefore, the system should provide sufficient guidance and contextual information for a new user to complete the consulting engagement lifecycle without requiring prior knowledge of IBM's consulting processes.

2.4 Operating Environment

Add in later

2.5 Design and Implementation

Add in later

2.6 Assumptions and Dependencies

AS-1: The system assumes that users have basic computer literacy and are able to interact with a desktop web application using standard input devices such as a keyboard and mouse.

AS-2: The system assumes that the target users have limited or no professional consulting experience. The application will therefore provide sufficient contextual information and guidance for users to progress through the core consulting lifecycle.

AS-3: The consulting mechanics, lead evaluation criteria, outreach practices, meeting behaviours, and engagement outcomes are dependent on information obtained from IBM consultants and project stakeholders. The accuracy of these mechanics is therefore dependent on the availability and quality of stakeholder input.

AS-4: The system assumes that the required LLM services are available when AI powered client interactions are initiated.

AS-5: The system assumes that users have a stable internet connection when accessing functionality that requires communication with external services.

AS-6: The system assumes that sufficient persona information can be defined for each simulated client, including their personality, objectives, concerns, knowledge and decision making behaviour.

AS-7: The system assumes that a user will complete the simulation within a session or within the persistence capabilities implemented by the project. The system is not intended to function as a continuously running multiplayer environment.

AS-8: The application is intended as a training simulation and does not represent actual IBM clients, engagements, contracts, or business decisions.

2.6 Edge Cases and Exceptional Scenarios

EC-1: If the user attempts to submit an empty cold outreach message, the system shall prevent submission and prompt the user to enter content.

EC-2: If the user exceeds the maximum message length for an outreach email or client response, the system shall notify the user and require the message to be shortened before submission.

EC-3: If the user enters inappropriate or unrelated text during an AI client conversation, the system shall maintain the client's professional persona and provide an appropriate in-context response.

EC-4: If the AI service is temporarily unavailable or fails to generate a response, the system shall display an error message and allow the user to retry the interaction.

EC-5: If an AI response is delayed, the system shall display a loading indicator and prevent duplicate message submissions until processing is complete.

EC-6: If the user's internet connection is interrupted during an AI interaction, the system shall notify the user that the request could not be completed and allow the interaction to be retried.

EC-7: If the user attempts to skip a required stage of the consulting lifecycle, the system shall prevent progression and indicate the remaining task that must be completed.

EC-8: If the user selects a low-quality or unsuitable lead, the system shall allow the engagement to continue while reflecting the decision through reduced deal potential or relationship outcomes rather than immediately ending the simulation.

EC-9: If the user performs poorly throughout an engagement, the system shall allow the engagement to conclude naturally and provide constructive feedback instead of forcing a successful outcome.

EC-10: If the user refreshes or accidentally closes the application during an active session, the system shall preserve the current game state where session persistence is supported and allow the user to resume the simulation.

3. Functional Requirements

3.1 User Access and Onboarding

FR3.1.1: The system shall allow a user to start a new simulation session. (H)

FR3.1.2: The system shall provide an onboarding/tutorial sequence explaining the objective of the simulation and the basic consulting gameplay mechanics. (H)

FR3.1.3: The system shall provide contextual instructions to guide users through each stage of the consulting lifecycle. (H)

FR3.1.4: The system shall allow the user to continue an active simulation session. (H)

FR3.1.5: The system shall display the user's current consulting level, portfolio progress, and active engagement status. (M)

3.2 Consulting World and Navigation

FR3.2.1: The system shall provide a 2D interactive environment representing the user's consulting workspace and relevant business locations. (H)

FR3.2.2: The system shall allow the user to navigate between available locations within the simulation environment. (H)

FR3.2.3: The system shall display available activities and interactions at each accessible location. (H)

FR3.2.4: The system shall unlock relevant locations, activities, and engagements based on the user's progression through the simulation. (M)

3.3 Lead Discovery

FR3.3.1: The system shall provide the user with a selection of potential business leads. (H)

FR3.3.2: The system shall provide relevant information about each lead to support the user's evaluation of the opportunity. (H)

FR3.3.3: The system shall allow the user to investigate and evaluate available leads. (H)

FR3.3.4: The system shall allow the user to select a lead to pursue. (H)

FR3.3.5: The system shall calculate or assign a lead score based on criteria established through consultation with IBM consultants. (H)

FR3.3.6: The system shall provide feedback on the suitability of the user's selected lead. (M)

3.4 Cold Outreach

FR3.4.1: The system shall allow the user to select an outreach method for a chosen lead. (H)

FR3.4.2: The system shall allow the user to compose an outreach message to the selected lead. (H)

FR3.4.3: The system shall allow the user to review their outreach message before submitting it. (H)

FR3.4.4: The system shall submit the user's outreach message to the selected lead. (H)

FR3.4.5: The system shall evaluate the effectiveness of the user's outreach based on predefined consulting criteria. (H)

FR3.4.6: The system shall provide a response from the lead based on the user's outreach. (H)

FR3.4.7: The system shall update the engagement state based on the outcome of the outreach. (H)

3.5 Meeting Preparation

FR3.5.1: The system shall provide the user with relevant information about the client before a scheduled meeting. (H)

FR3.5.2: The system shall allow the user to review information about the client's organisation, industry, and identified business needs. (H)

FR3.5.3: The system shall allow the user to establish objectives for the client meeting. (H)

FR3.5.4: The system shall allow the user to prepare questions or discussion topics for the client meeting. (H)

FR3.5.5: The system shall provide feedback on the user's meeting preparation. (M)

FR3.5.6: The system shall use relevant information from the user's preparation to inform the subsequent client interaction. (H)

3.6 AI powered Client Interaction

FR3.6.1: The system shall provide at least one LLM-powered client persona for user interaction. (H)

FR3.6.2: The system shall allow the user to communicate with the client persona using natural language. (H)

FR3.6.3: The system shall generate client responses dynamically based on the user's input. (H)

FR3.6.4: The client persona shall respond according to its assigned personality, goals, concerns, and available knowledge. (H)

FR3.6.5: The client persona shall be capable of generating questions, objections, requests for clarification, and other responses relevant to the simulated engagement. (H)

FR3.6.6: The system shall maintain the context of the current client conversation throughout the meeting. (H)

FR3.6.7: The system shall prevent the client persona from accessing or generating information outside of its defined simulation context where applicable. (M)

FR3.6.8: The system shall provide the user with an indication when an AI-generated response is being processed. (M)

3.7 Client Relationship and Meeting Outcomes

FR3.7.1: The system shall maintain a relationship state between the user and each active client. (H)

FR3.7.2: The system shall update the client's relationship state based on relevant user actions and interaction outcomes. (H)

FR3.7.3: The system shall track the outcome of client meetings. (H)

FR3.7.4: The system shall evaluate the user's meeting performance against defined consulting criteria. (H)

FR3.7.5: The system shall provide feedback on the user's meeting performance after the interaction. (H)

FR3.7.6: The system shall allow meeting outcomes to affect the progression of the current engagement. (H)

3.8 Proposal and Contract

FR3.8.1: The system shall allow the user to develop a proposed engagement based on information obtained during the client interaction. (H)

FR3.8.2: The system shall allow the user to define relevant aspects of the proposed engagement. (H)

FR3.8.3: The system shall present the proposed engagement to the simulated client. (H)

FR3.8.4: The client persona shall respond to the proposed engagement based on its defined objectives, concerns, and decision making criteria. (H)

FR3.8.5: The system shall allow the user to respond to client objections or negotiation points. (H)

FR3.8.6: The system shall determine an engagement outcome based on the user's interactions and decisions. (H)

FR3.8.7: The system shall record whether the engagement resulted in a successful contract, unsuccessful contract, or other defined outcome. (H)

3.9 Portfolio and Progress Tracking

FR3.9.1: The system shall maintain the user's portfolio throughout a simulation session. (H)

FR3.9.2: The system shall track the number of contracts closed by the user. (H)

FR3.9.3: The system shall track the outcome and quality of completed engagements. (H)

FR3.9.4: The system shall track the relationship health associated with active or completed engagements. (H)

FR3.9.5: The system shall update portfolio statistics following the completion of an engagement. (H)

FR3.9.6: The system shall display the user's portfolio progress within the simulation. (H)

FR3.9.7: The system shall retain relevant game state throughout the user's session. (H)

3.10 Feedback and Learning

FR3.11.1: The system shall provide feedback following significant consulting activities. (H)

FR3.11.2: The system shall identify strengths and areas for improvement based on the user's actions. (H)

FR3.11.3: The system shall provide an outcome summary after the completion of an engagement. (H)

FR3.11.4: The system shall provide the user with an explanation of factors that contributed to the engagement outcome where appropriate. (H)

FR3.11.5: The system shall allow the user to review their performance within the completed engagement. (M)

FR3.11.6: The system shall use performance outcomes to inform the user's progression through subsequent simulation content. (M)

3.11 Client Personas

FR3.11.1: The system shall support multiple client personas. (M)

FR3.11.2: Each client persona shall have defined characteristics, including personality, objectives, concerns, and decision making behaviour. (M)

FR3.11.3: The system shall provide different client scenarios with varying levels of difficulty. (M)

FR3.11.4: The system shall select or assign an appropriate client persona to an engagement. (M)

FR3.11.5: The system shall ensure that client responses remain consistent with the selected persona throughout an engagement. (H)

3.12 Simulation Progression

FR3.12.1: The system shall track the user's progression through the consulting lifecycle. (H)

FR3.12.2: The system shall unlock subsequent stages of an engagement when the required preceding stage has been completed. (H)

FR3.12.3: The system shall allow engagement outcomes to affect the user's progression. (M)

FR3.12.4: The system shall provide the user with an indication of their current progression and objectives. (H)

FR3.12.5: The system shall allow the user to complete a full consulting engagement from lead identification through contract outcome. (H)